

COMPSCI 103. Advanced Programming and Problem Solving with C++. 4 Units.

Advanced programming language concepts for more complex, higher performance software design. Builds depth of programming skills in C++ as a foundation for upper-division courses and projects. Focuses on strengthening programming, debugging, and problem solving skills.

Prerequisite: **I&C SCI 45C**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 111. Digital Image Processing. 4 Units.

Introduction to the fundamental concepts of digital signal and image processing as applicable in areas such as multimedia, graphics, AI, data mining, databases, vision, or video games. Topics include image representation, space- and frequency-domain transformations, filters, segmentation, and compression.

Prerequisite: **I&C SCI 46** with a minimum grade of C and **I&C SCI 6D** with a minimum grade of C and (**MATH 3A** with a minimum grade of C or **I&C SCI 6N** with a minimum grade of C).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 112. Computer Graphics. 4 Units.

Introduction to the fundamental principles of 3D computer graphics including polygonal modeling, geometric transformations, visibility algorithms, illumination models, texturing, and rasterization. Use of an independently-learned 3D graphics API to implement these techniques.

Prerequisite: **I&C SCI 46** with a minimum grade of C and (**MATH 3A** with a minimum grade of C or **I&C SCI 6N** with a minimum grade of C).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 114. Projects in Advanced 3D Computer Graphics. 4 Units.

Projects in advanced 3D graphics such as illumination, geometric modeling, visualization, and animation. Topics include physically based and global illumination, solid modeling, curved surfaces, multiresolution modeling, image-based rendering, basic concepts of animation, and scientific visualization.

Prerequisite: [COMPSCI 112](#) and [I&C SCI 45C](#) with a minimum grade of C. Recommended: [COMPSCI 161](#) or [COMPSCI 164](#) or [COMPSCI 165](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 115. Computer Simulation. 4 Units.

Discrete event-driven simulation; continuous system simulation; basic probability as pertaining to input distributions and output analysis; stochastic and deterministic simulation; static and dynamic system simulation.

Prerequisite: [I&C SCI 6B](#) with a minimum grade of C and ([I&C SCI 6N](#) with a minimum grade of C or [MATH 3A](#) with a minimum grade of C) and ([STATS 67](#) with a minimum grade of C or ([STATS 7](#) with a minimum grade of C or AP Statistics with a minimum score of 3) and [STATS 120A](#) with a minimum grade of C) and [I&C SCI 51](#) with a minimum grade of C and ([SWE 43](#) with a minimum grade of C or [IN4MATX 43](#) with a minimum grade of C).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 116. Computational Photography and Vision. 4 Units.

Introduces the problems of computer vision through the application of computational photography. Specific topics include photo-editing (image warping, compositing, hole filling), panoramic image stitching, and face detection for digital photographs.

Prerequisite: [I&C SCI 6D](#) with a minimum grade of C and ([MATH 3A](#) with a minimum grade of C or [I&C SCI 6N](#) with a minimum grade of C) and ([MATH 2B](#) with a minimum grade of C or AP Calculus BC with a minimum score of 4 or AP Calculus AB with a minimum score of 4) and [I&C SCI 46](#) with a minimum grade of C.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 117. Project in Computer Vision. 4 Units.

Students undertake construction of a computer vision system. Topics include automatically building 3D models from photographs, searching photo collections, robot navigation, and human motion tracking.

Prerequisite: I&C SCI 6D with a minimum grade of C and (MATH 3A with a minimum grade of C or I&C SCI 6N with a minimum grade of C) and (MATH 2B with a minimum grade of C or AP Calculus BC with a minimum score of 4) and I&C SCI 46 with a minimum grade of C and (COMPSCI 112 or COMPSCI 116 or COMPSCI 171 or COMPSCI 178).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 118. Introduction to Virtual Reality. 4 Units.

Introduces virtual reality systems and software. Topics include computer graphics pipeline, human visual system and perception, head mounted displays, image formation, sensors and tracking, and 3D sound. Students use a popular cross-platform game engine to create VR environments.

Prerequisite: (I&C SCI 6N or MATH 3A) and I&C SCI 33.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 121. Information Retrieval. 4 Units.

An introduction to information retrieval including indexing, retrieval, classifying, and clustering text and multimedia documents.

Prerequisite: (I&C SCI 45C with a minimum grade of C or I&C SCI 45J with a minimum grade of C) and (STATS 7 or STATS 67 or MGMT 7 or AP Statistics with a minimum score of 3).

Same as SWE 141

Restrictions: School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 122A. Introduction to Data Management. 4 Units.

Introduction to the design of databases and the use of database management systems (DBMS) for applications. Topics include entity-relationship modeling for design, relational data model, relational algebra, relational design theory, and Structured Query Language (SQL) programming.

Prerequisite: **I&C SCI 33** with a minimum grade of C or **EECS 114**.

Same as **EECS 116**

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 122B. Project in Databases and Web Applications. 4 Units.

Introduces students to advanced database technologies and web applications. Topics include database connectivity (ODBC/JDBC), extending databases using stored procedures, database administration, web servers, web programming languages (Java servlets, XML, Ajax, and mobile platforms).

Prerequisite: (**COMPSCI 122A** or **EECS 116**) and (**I&C SCI 45J** or AP Computer Science A with a minimum score of 4).

Overlaps with **COMPSCI 137**, IN4MATX 124.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 122C. Principles of Data Management. 4 Units.

Covers fundamental principles underlying data management systems. Content includes key techniques including storage management, buffer management, record-oriented file system, access methods, query optimization, and query processing.

Prerequisite: **COMPSCI 122A** and (**I&C SCI 53** or **COMPSCI 143A**).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: **COMPSCI 222**

COMPSCI 122D. Beyond SQL Data Management. 4 Units.

Survey of modern data management and analysis technologies beyond relational (SQL) database management. Topics include semistructured data, NoSQL databases, big data analysis, and text search/query engines. Additional topics vary and may include data frames, streaming data, and/or time-series data.

Prerequisite: **I&C SCI 46** and (**I&C SCI 51** or **EECS 31**) and (**COMPSCI 122A** or **EECS 116**).

Restrictions: Computer Science majors have the first consideration for enrollment. Computer Science and Engineering and all School of ICS majors have second right of consideration.

COMPSCI 125. Next Generation Search Systems. 4 Units.

Discusses concepts and techniques related to all aspects of search systems. After considering basic search technology and the state-of-art systems, rapidly developing techniques for multimedia search, local search, event-search, and video-on-demand are explored.

Prerequisite: (**I&C SCI 45C** with a minimum grade of C or **I&C SCI 45J** with a minimum grade of C) and (**STATS 7** or **STATS 67** or AP Statistics with a minimum score of 3).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 130. Introduction to Security and Privacy. 4 Units.

Introductory course on security mindset and basic principles/practices of software and systems security. Covers the foundations of building, using, and managing secure systems. Topics include standard security principles, threats, and defenses for real-world systems, including web, mobile, and IoT security.

Prerequisite: **I&C SCI 46** with a minimum grade of C.

Restrictions: Computer Science majors, Computer Science and Engineering majors, and School of Information and Computer Sciences students only.

COMPSCI 131. Parallel and Distributed Computing. 4 Units.

Parallel and distributed computer systems. Parallel programming models. Common parallel and distributed programming issues. Specific topics include parallel programming, performance models, coordination and synchronization, consistency and replication, transactions, fault tolerance.

Prerequisite: [I&C SCI 53](#) or [COMPSCI 143A](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 132. Computer Networks. 4 Units.

Computer network architectures, protocols, and applications. Internet congestion control, addressing, and routing. Local area networks. Multimedia networking.

Prerequisite: [EECS 55](#) or [STATS 67](#).

Same as [EECS 148](#)

Restrictions: Computer Engineering majors and Computer Science and Engineering majors have the first consideration for enrollment.

COMPSCI 133. Advanced Computer Networks. 4 Units.

Fundamental principles in computer networks are applied to obtain practical experience and skills necessary for designing and implementing computer networks, protocols, and network applications. Various network design techniques, simulation techniques, and UNIX network programming are covered.

Prerequisite: [COMPSCI 132](#) or [EECS 148](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 134. Computer and Network Security. 4 Units.

Overview of modern computer and networks security, attacks, and countermeasures. Authentication, identification, data secrecy, data integrity, authorization, access control, computer viruses, network security. Also covers secure e-commerce and applications of public key methods, digital certificates, and credentials.

Prerequisite: [COMPSCI 161](#) and [I&C SCI 53](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 137. Internet Applications Engineering. 4 Units.

Concepts in Internet applications engineering with emphasis on the Web. Peer-to-Peer and Interoperability. Topics include HTTP and REST, Remote Procedure/Method Calls, Web Services, data representations, content distribution networks, identity management, relevant W3C/IETF standards, and relevant new large-scale computing styles.

Prerequisite: ([COMPSCI 132](#) or [EECS 148](#)) and [I&C SCI 45J](#).

Same as [SWE 124](#)

Overlaps with [COMPSCI 122B](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 141. Concepts in Programming Languages. 4 Units.

In-depth study of several contemporary programming languages stressing variety in data structures, operations, notation, and control. Examination of different programming paradigms, such as logic programming, functional programming and object-oriented programming; implementation strategies, programming environments, and programming style.

Prerequisite: ([I&C SCI 51](#) with a minimum grade of C or [EECS 31](#) with a minimum grade of C) and [I&C SCI 46](#) with a minimum grade of C.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students only.

COMPSCI 142A. Compilers and Interpreters. 4 Units.

Introduction to the theory of programming language processors covering lexical analysis, syntax analysis, semantic analysis, intermediate representations, code generation, optimization, interpretation, and run-time support.

Prerequisite: (I&C SCI 46 or EECS 114) and (I&C SCI 51 or EECS 31). Recommended: COMPSCI 141 or IN4MATX 101 or I&C SCI 45J or EECS 40

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students only.

COMPSCI 142B. Project in Compilers and Interpreters. 4 Units.

Project course which provides working laboratory experience in construction and behavior of compilers and interpreters. Students build actual language processors and perform experiments which reveal their behaviors.

Prerequisite: COMPSCI 142A.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 143A. Principles of Operating Systems. 4 Units.

Principles and concepts of process and resource management, especially as seen in operating systems. Processes, memory management, protection, scheduling, file systems, and I/O systems are covered. Concepts illustrated in the context of several well-known systems.

Prerequisite: (I&C SCI 46 or CSE 46) and (I&C SCI 51 or EECS 31 or CSE 31).

Overlaps with EECS 111.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 143B. Project in Operating System Organization. 4 Units.

Detailed specification and design of critical components of an actual operating system including a memory manager, a process server, and a file/IO subsystem. Hardware/software tradeoffs. Emphasis on logical organization of system and communication.

Prerequisite: [COMPSCI 143A](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 145. Embedded Software. 6 Units.

Principles of embedded computing systems: embedded systems architecture, hardware/software components, system software and interfacing, real-time operating systems, hardware/software co-development, and communication issues. Examples of embedded computing in real-world application domains. Simple programming using an embedded systems development environment.

Prerequisite: [I&C SCI 46](#) and ([I&C SCI 51](#) or [EECS 31](#)).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 146. Programming in Multitasking Operating Systems. 4 Units.

User- and systems-level programming of modern Internet-connected, multi-user, multitasking operating systems. Shells, scripting, filters, pipelines, programmability, extensibility, concurrency, inter-process communication. Concrete examples of a modern operating system (such as, but not necessarily, Unix programmed in C) are used.

Prerequisite: [I&C SCI 46](#) with a minimum grade of C and [I&C SCI 51](#) with a minimum grade of C. Recommended: [COMPSCI 143A](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 147. Internet of Things (IoT) Software and Systems. 4 Units.

Introduction to the Internet of Things (IoT) from a systems and software perspective. IoT ecosystem including sensors, embedded CPUs, networking protocols, software, cloud services, and security and privacy requirements. IoT use cases, system design, and programming project.

Prerequisite: **I&C SCI 33**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 151. Digital Logic Design. 4 Units.

Boolean algebra. Design/analysis of combinational and sequential systems using SSI/MSI/LSI modules. Number systems. Error detecting and correction codes. Arithmetic algorithms. Hardware/firmware implementation of algorithms.

Prerequisite: **I&C SCI 33** with a minimum grade of C and **I&C SCI 51** with a minimum grade of C and **I&C SCI 6B** and **I&C SCI 6D**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 152. Computer Architecture. 4 Units.

Design of computer elements; ALU, control unit, and arithmetic circuits. Memory hierarchy and organization. Caches. Function unit sharing and pipelining. I/O and interrupt processing. RTL and behavioral modeling using hardware description languages. Microprocessor organization and implementation techniques.

Prerequisite: **I&C SCI 51**.

Overlaps with I&C SCI 160, **EECS 112**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 154. Computer Design Laboratory. 4 Units.

Underlying primitives of computer instruction sets. Principles of microprogramming. Microprogramming. Microprograms written for one or more systems. Typical microprogramming applications discussed and implemented or simulated.

Prerequisite: [COMPSCI 151](#) or [COMPSCI 152](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 161. Design and Analysis of Algorithms. 4 Units.

Techniques for efficient algorithm design, including divide-and-conquer and dynamic programming, and time/space analysis. Fast algorithms for problems applicable to networks, computer games, and scientific computing, such as sorting, shortest paths, minimum spanning trees, network flow, and pattern matching.

Prerequisite: [I&C SCI 46](#) with a minimum grade of C and [I&C SCI 6B](#) and [I&C SCI 6D](#) and ([MATH 2B](#) or AP Calculus BC with a minimum score of 4).

Restrictions: Computer Science and Engineering majors, Computer Science majors, Data Science majors, Software Engineering majors, and School of Information and Computer Sciences students only. Business Information Management, Computer Game Science, and Informatics majors have second right of consideration.

COMPSCI 162. Formal Languages and Automata. 4 Units.

Formal aspects of describing and recognizing languages by grammars and automata. Parsing regular and context-free languages. Ambiguity, nondeterminism. Elements of computability; Turing machines, random access machines, undecidable problems, NP-completeness.

Prerequisite: [I&C SCI 46](#) with a minimum grade of C and ([MATH 2B](#) or AP Calculus BC with a minimum score of 4) and [I&C SCI 6B](#) and [I&C SCI 6D](#).

Same as [LSCI 102](#)

Restrictions: Cognitive Sciences majors, Computer Science and Engineering majors, Language Science majors, and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 163. Graph Algorithms. 4 Units.

Algorithms for solving fundamental problems in graph theory. Graph representations, graph traversal, network flow, connectivity, graph layout, matching problems.

Prerequisite: [COMPSCI 161](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: [COMPSCI 265](#)

COMPSCI 164. Computational Geometry. 4 Units.

Algorithms and data structures for computational geometry and geometric modeling, with applications to computer vision and graphics. Topics: convex hulls, Voronoi diagrams, algorithms for triangulation, motion planning, and data structures for geometric searching and modeling of 2D and 3D

Prerequisite: [COMPSCI 161](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: [COMPSCI 266](#)

COMPSCI 165. Project in Algorithms and Data Structures. 4 Units.

Design, implementation, execution, and analysis of algorithms for problems such as sorting, searching, data compression, and data encryption. Time-space-structure trade-offs.

Prerequisite: [COMPSCI 161](#). Recommended: [I&C SCI 45C](#).

Restrictions: Computer Science majors and School of Information and Computer Sciences students have the first consideration for enrollment. Computer Science and Engineering majors and all School of ICS majors have second right of consideration.

COMPSCI 166. Quantum Computation and Information. 4 Units.

Basic models for quantum computation and their foundations in quantum mechanics. Quantum complexity classes and quantum algorithms, including algorithms for factoring and quantum simulation. Introduction to quantum information theory and quantum entanglement.

Prerequisite: (I&C SCI 6N with a minimum grade of C+ or MATH 3A with a minimum grade of C+) and (COMPSCI 161 or PHYSICS 113A).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 167. Introduction to Applied Cryptography. 4 Units.

An introduction to the essential aspects of applied cryptography, as it is used in practice. Topics include classical cryptography, block ciphers, stream ciphers, public-key cryptography, digital signatures, one-way hash functions, basic cryptographic protocols, and digital certificates and credentials.

Prerequisite: COMPSCI 161.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 169. Introduction to Optimization. 4 Units.

A broad introduction to optimization. Unconstrained and constrained optimization. Linear and integer programming. Combinatorial and geometric optimization. AI-assisted optimization and optimization applications to AI.

Prerequisite: (I&C SCI 6N or MATH 3A) and (MATH 2B or AP Calculus BC with a minimum score of 4) and COMPSCI 161.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: COMPSCI 268

COMPSCI 171. Introduction to Artificial Intelligence. 4 Units.

Different means of representing knowledge and uses of representations in heuristic problem solving. Representations considered include predicate logic, semantic nets, procedural representations, natural language grammars, and search trees.

Prerequisite: **I&C SCI 46** and (**MATH 2B** or AP Calculus BC with a minimum score of 4) and **STATS 67** or ((**STATS 7** or AP Statistics with a minimum score of 3) and **STATS 120A**).

Overlaps with **EECS 118**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 172B. Neural Networks and Deep Learning . 4 Units.

Neural network and deep learning from multiple perspectives. Theory of parallel distributed processing systems, algorithmic approaches for learning from data in various manners, applications to difficult problems in AI from computer vision, to natural language understanding, to bioinformatics and chemoinformatics.

Prerequisite: (**STATS 120A** and **STATS 120B**) or **MATH 121A** or **COMPSCI 178** or **COMPSCI 273A**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: **COMPSCI 274C**

COMPSCI 172C. Artificial Intelligence Frontiers: Technical, Ethical, and Societal. 4 Units.

Explores the frontiers of artificial intelligence and related technologies with a focus on the underlying ethical, legal, and societal challenges and opportunities they create. Encourages critical thinking about these issues.

Prerequisite: **COMPSCI 171** and **COMPSCI 172B**. Recommended: **COMPSCI 178**

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: **COMPSCI 274D**

COMPSCI 175. Project in Artificial Intelligence. 4 Units.

Construction of a working artificial intelligence system. Evaluation of capabilities of the system including impact of knowledge representation.

Prerequisite: **COMPSCI 171** and **COMPSCI 178**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 177. Applications of Probability in Computer Science. 4 Units.

Application of probability to real-world problems in computer science. Typical topics include analysis of algorithms and graphs, probabilistic language models, network traffic modeling, data compression, and reliability modeling.

Prerequisite: (**MATH 2B** or AP Calculus BC with a minimum score of 4) and **I&C SCI 6B** and **I&C SCI 6D** and (**MATH 3A** or **I&C SCI 6N**) and **STATS 67**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 178. Machine Learning and Data-Mining. 4 Units.

Introduction to principles of machine learning and data-mining applied to real-world datasets. Typical applications include spam filtering, object recognition, and credit scoring.

Prerequisite: **I&C SCI 6B** and **I&C SCI 6D** and (**I&C SCI 6N** or **MATH 3A**) and (**MATH 2B** or AP Calculus BC with a minimum score of 4) and **STATS 67** or ((**STATS 7** or AP Statistics with a minimum score of 3) and **STATS 120A**).

Overlaps with **EECS 125**.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 179. Algorithms for Probabilistic and Deterministic Graphical Models. 4 Units.

Graphical model techniques dealing with probabilistic and deterministic knowledge representations. Focuses on graphical models such as constraint networks, Bayesian networks, and Markov networks that have become a central paradigm for knowledge representation and reasoning in AI and general computer science.

Prerequisite: [COMPSCI 171](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 180A. Project in Computer Science. 4 Units.

Students to solve a substantial real-world problem with knowledge gained from many areas in computer science. Project has a focus on computer science but can overlap with neighbor disciplines.

Prerequisite: [I&C SCI 46](#).

Restrictions: School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 180B. Project in Computer Science. 4 Units.

Students to solve a substantial real-world problem with knowledge gained from many areas in computer science. Project has a focus on computer science but can overlap with neighbor disciplines.

Prerequisite: [COMPSCI 180A](#). In Progress (IP) grade for [COMPSCI 180A](#) is also accepted.

Restrictions: School of Information and Computer Sciences students have the first consideration for enrollment.

COMPSCI 183. Introduction to Computational Biology. 4 Units.

The use of theories and methods based on computer science, mathematics, and physics in molecular biology and biochemistry. Basics in biomolecular modeling. Analysis of sequence and structural data of biomolecules. Analysis of biomolecular functions.

Prerequisite: [MATH 2D](#) or [MATH 3A](#) or [STATS 7](#) or [STATS 8](#).

Same as [BIO SCI M123](#), [BME 132](#)

Concurrent: [MOL BIO 223](#), [BME 232](#)

COMPSCI 184A. Artificial Intelligence in Biology and Medicine. 4 Units.

Introduction to computational methods in molecular biology, aimed at those interested in learning about this interdisciplinary area. Covers computational approaches to understanding and predicting the structure, function, interactions, and evolution of DNA, RNA, proteins, and related molecules and processes.

Prerequisite: [I&C SCI 6N](#) or [MATH 3A](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: [COMPSCI 284A](#)

COMPSCI 184C. Computational Systems Biology. 4 Units.

Computational inference and modeling of gene regulation networks, signal transduction pathways, and the effects of regulatory networks in cellular processes, development, and disease. Introduction of required mathematical, computational, and data handling tools.

Prerequisite: [COMPSCI 184A](#).

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.

Concurrent: [COMPSCI 284C](#)

COMPSCI 189. Project in Bioinformatics. 4 Units.

Teaches problem definition and analysis, data representation, algorithm design, component integration, solution validation, and testability with teams specifying, designing, building, and testing a solution to a bioinformatics problem. Lectures include engineering values, discussions, and ethical ramifications of biomedical computing issues.

Prerequisite: **COMPSCI 184A** with a minimum grade of C.

Restrictions: Computer Science and Engineering majors and School of Information and Computer Sciences students have the first consideration for enrollment.